

User Requirement Specification - Two-Tank Sequence Controller

Document	URS-001
Revision	A
Date	15 Jan 2026
Status	Released - baseline, subject to approved changes
Classification	Synthetic hackathon artifact

1. Purpose and Scope

The package controls a demonstration liquid-transfer process consisting of two tanks, three actuated isolation valves, two level measurements and three operator pushbuttons. The controller is expected to execute the normal sequence automatically once started and to support pause/resume and controlled shutdown commands.

2. System Identification

Project tag	Description	Common alias
TK-101	Tank 1	tank1 / T1
TK-102	Tank 2	tank2 / T2
XV-101	Inlet valve to Tank 1	valve1 / V1
XV-102	Transfer valve Tank 1 to Tank 2	valve2 / V2
XV-103	Tank 2 drain valve	valve3 / V3
PLC-101	Sequence controller	tankController

3. Functional Requirements

URS-F-001 - START shall initiate automatic operation when the system is in its initial condition.

URS-F-002 - Normal operation shall fill Tank 1 by opening V1.

URS-F-003 - V1 shall close when Tank 1 reaches the specified high-level limit.

URS-F-004 - After a 10 s waiting period, V2 shall open and transfer liquid from Tank 1 to Tank 2.

URS-F-005 - V2 shall close when Tank 1 reaches its minimum operating level.

URS-F-006 - After a 10 s waiting period, V3 shall open and drain Tank 2.

URS-F-007 - V3 shall close when Tank 2 reaches its minimum operating level.

URS-F-008 - The automatic process shall continue cyclically unless stopped or shut down.

URS-M-001 - STOP shall close all three valves and suspend automatic operation until another operator command is received.

URS-M-002 - START after STOP shall continue the process operation rather than force a new empty-cycle initialization.

URS-M-003 - SHUT shall empty both tanks by opening V2 and V3 and shall inhibit V1.

URS-M-004 - When shutdown emptying is achieved, V2 and V3 shall close and the controller shall return to the start configuration.

4. Initial Engineering Setpoints

The following values were agreed for the Rev A baseline. Approved engineering changes issued after this document take precedence.

Parameter	Rev A value	Unit	Comment
Tank 1 initial level	0.05	m	Demo initialization
Tank 2 initial level	0.05	m	Demo initialization
Tank 1 high limit	0.78	m	Subject to later controls tuning
Tank 1 low limit	0.05	m	Residual heel
Tank 2 low limit	0.05	m	Residual heel
Wait after T1 high	10	s	No valve open during wait
Wait after T1 low	10	s	No valve open during wait
Wait after T2 low	10	s	Before next automatic fill cycle

5. Interfaces

Level measurements LT-101 and LT-102 shall be available to PLC-101. START, STOP and SHUT are momentary commands. Valve commands are binary open/closed outputs. The supplier shall provide actuators that fail closed on loss of power.

6. Verification Intent

Sequence operation, stop/resume behavior and controlled shutdown shall be demonstrated in simulation. The exact demonstration command schedule will be issued in the validation procedure.